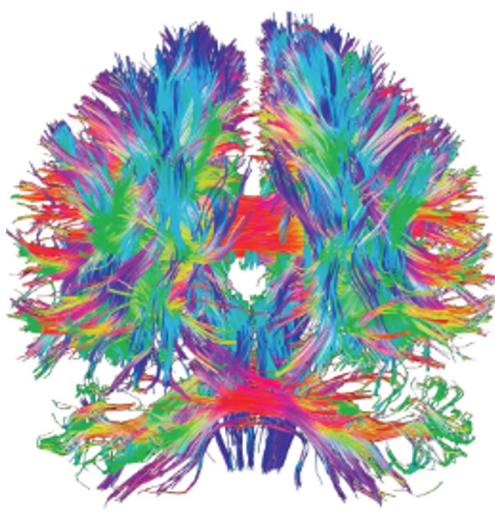




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Neuroinformatics

Detection of Cerebrovascular Diseases with in Vivo MRI and Deep Neural Networks

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Introduction

- **Cerebrovascular disease (CVD)** contributes to the risk of other diseases and can result in death¹
- Some vascular changes may go **undetected** on routine inspection of brain scans or require specialized imaging
- **Deep neural networks** may be able to detect CVD using **MRI** and learn about its global impact on brain structure
- We hypothesized there might be some alterations in brain structure that would be detectable through MRI in patients with CVD
- We aimed to develop a **Convolutional Neural Network (CNN)** to detect CVD and its biomarkers accurately from patients' MRI scans

Methods

Case-Control Matching

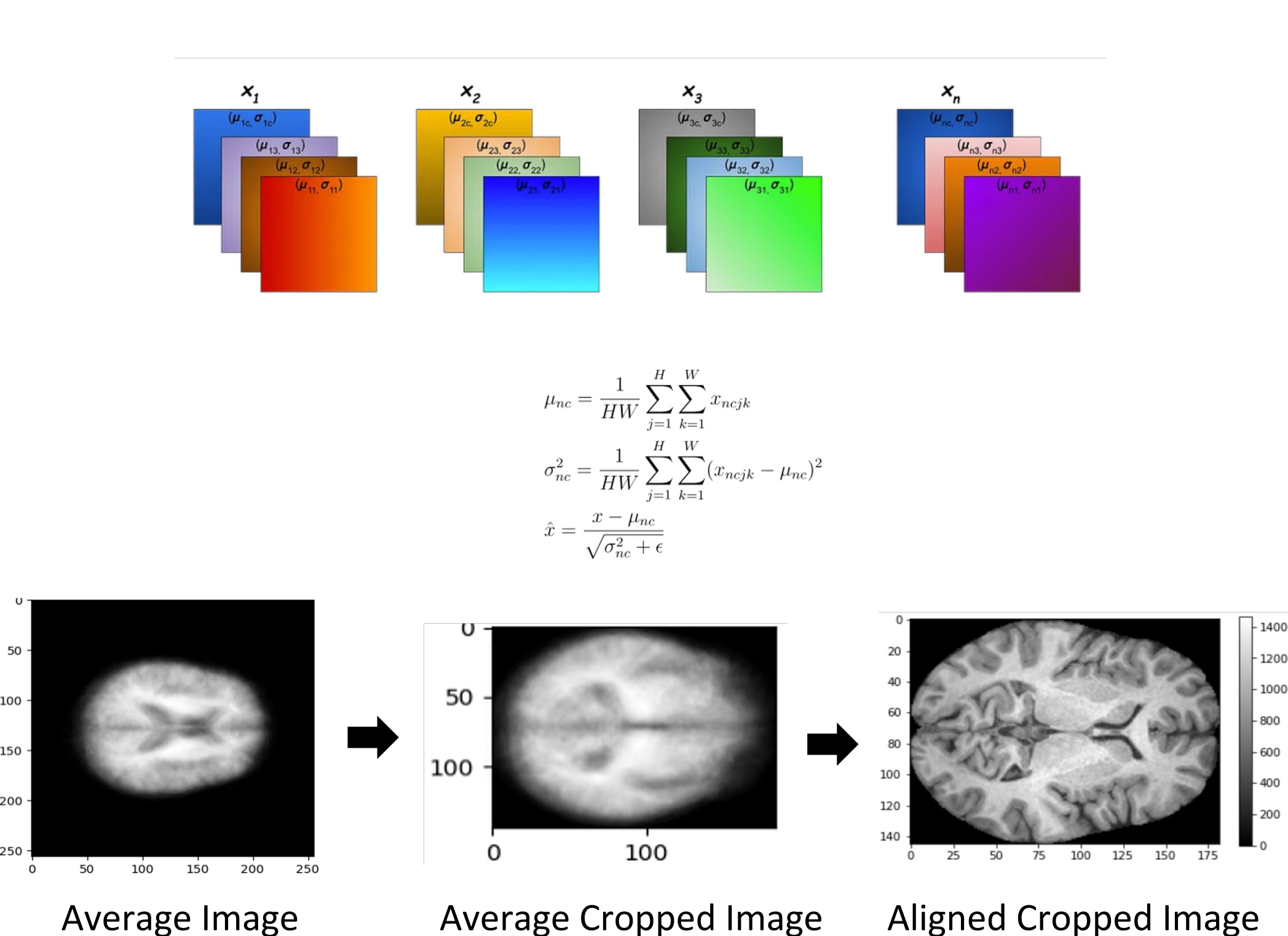
MRI data is from **UK Biobank (originally N = 500,000)**

We matched case and control with:

- **1:1 case - control ratio**
- 4 variables – **Age, BMI, Sex, Ethnicity**

There are **237** positive cases, and **237** negative cases in the matched dataset - **474 in total**, then split into training, validation, and testing with ratio of **7:2:1**

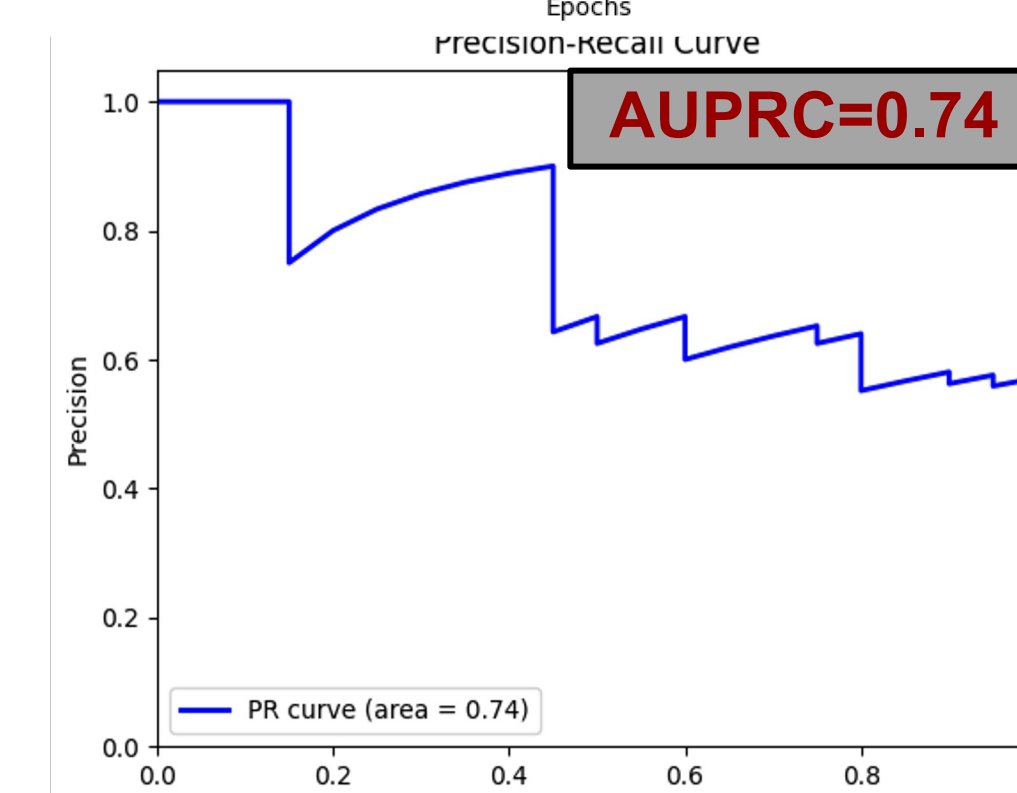
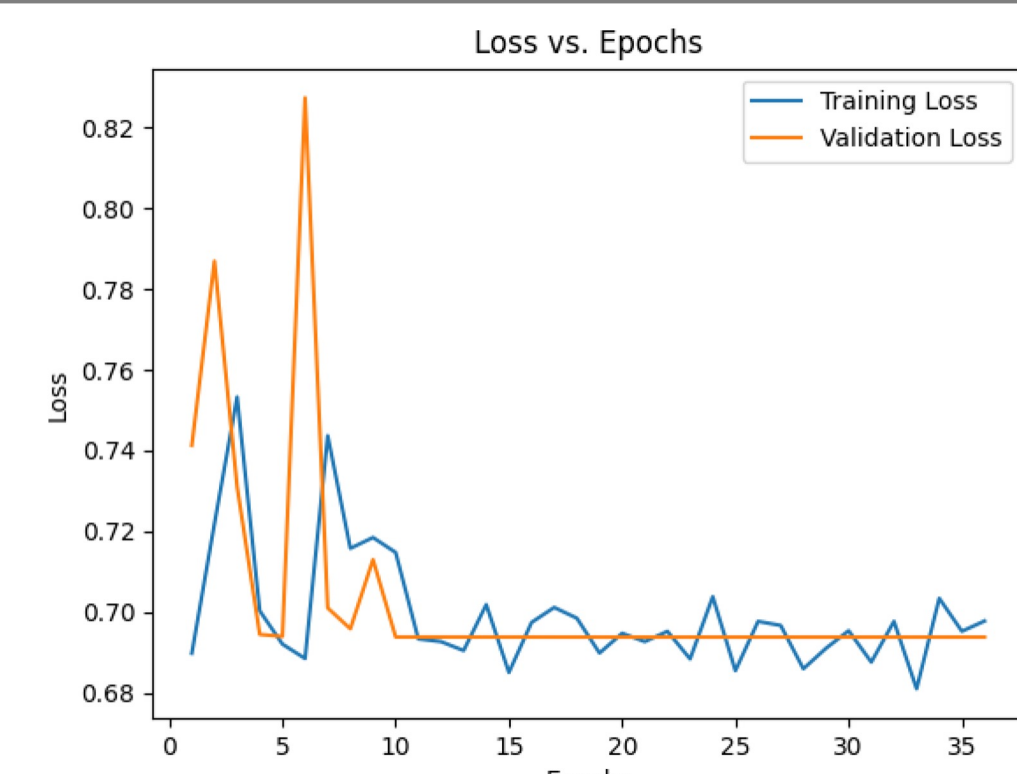
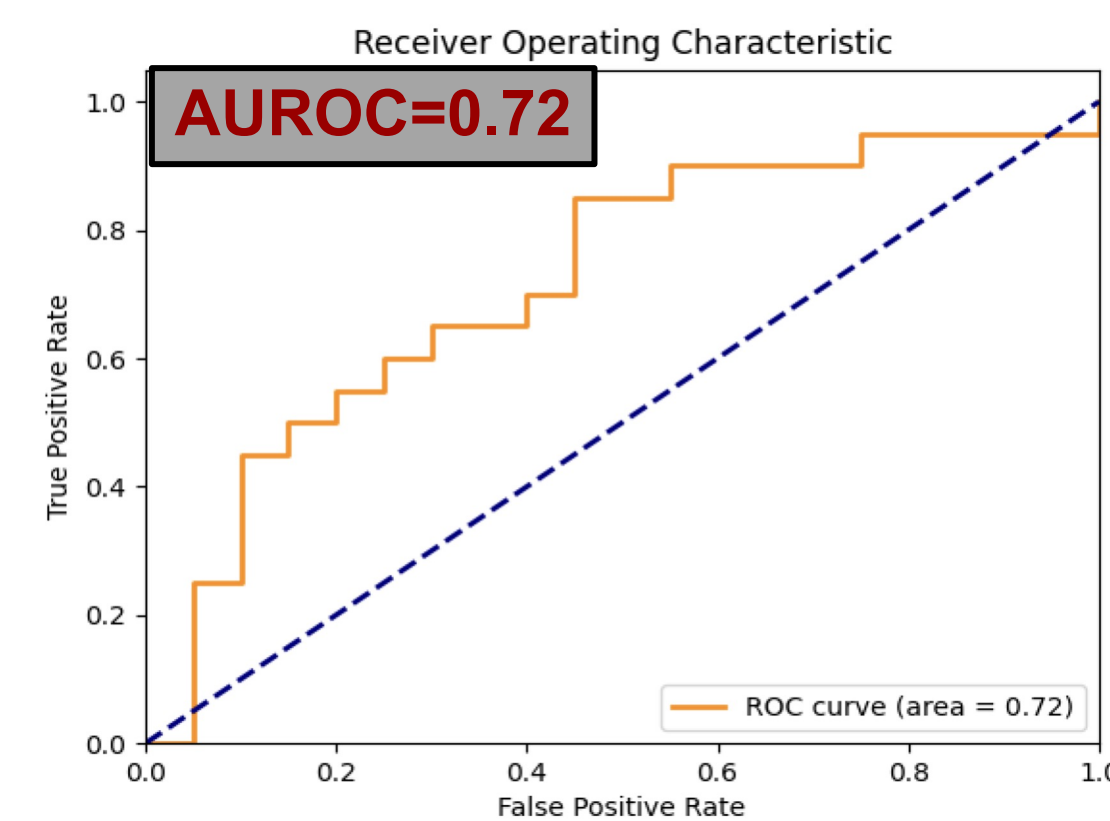
Convolutional Neural Network



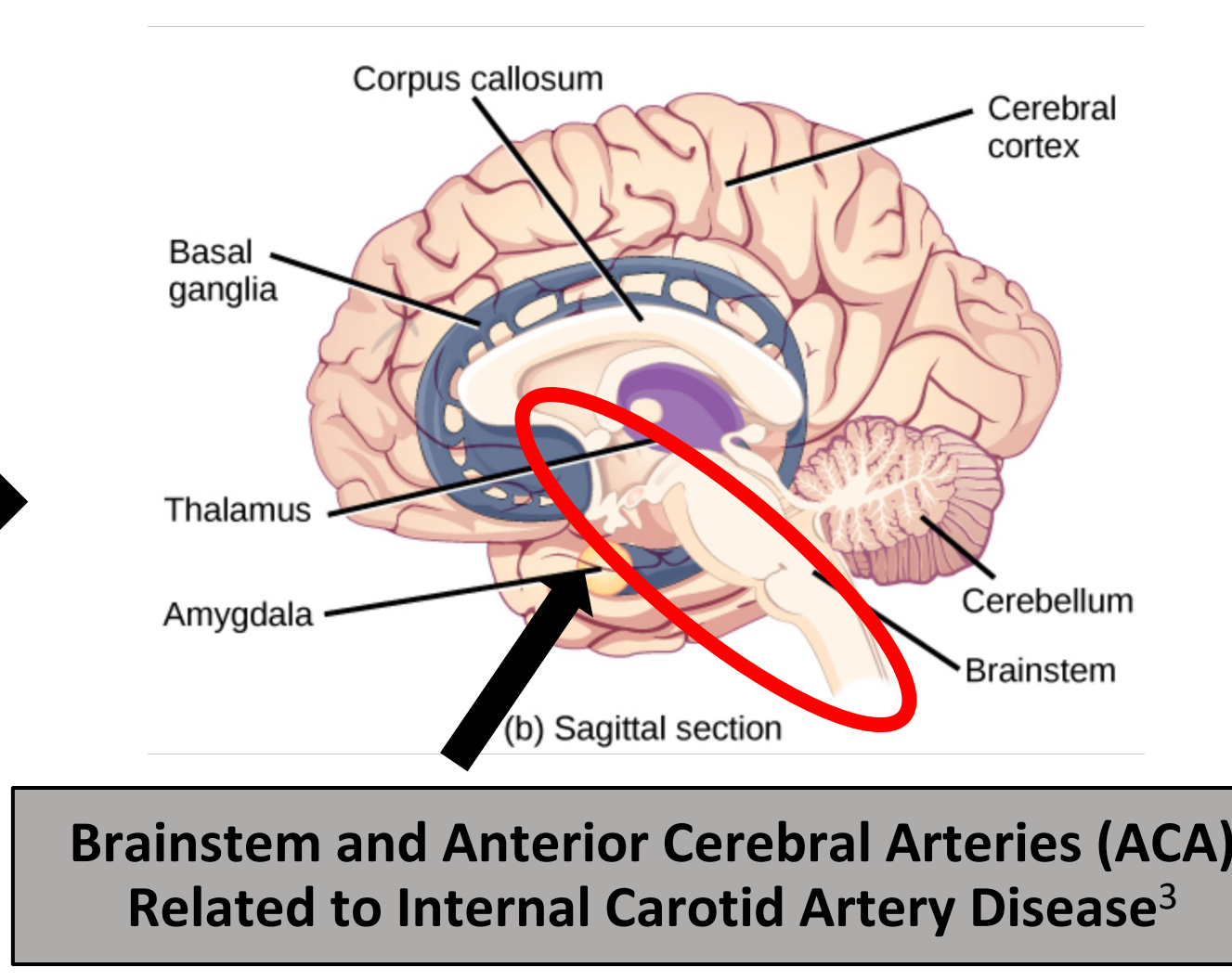
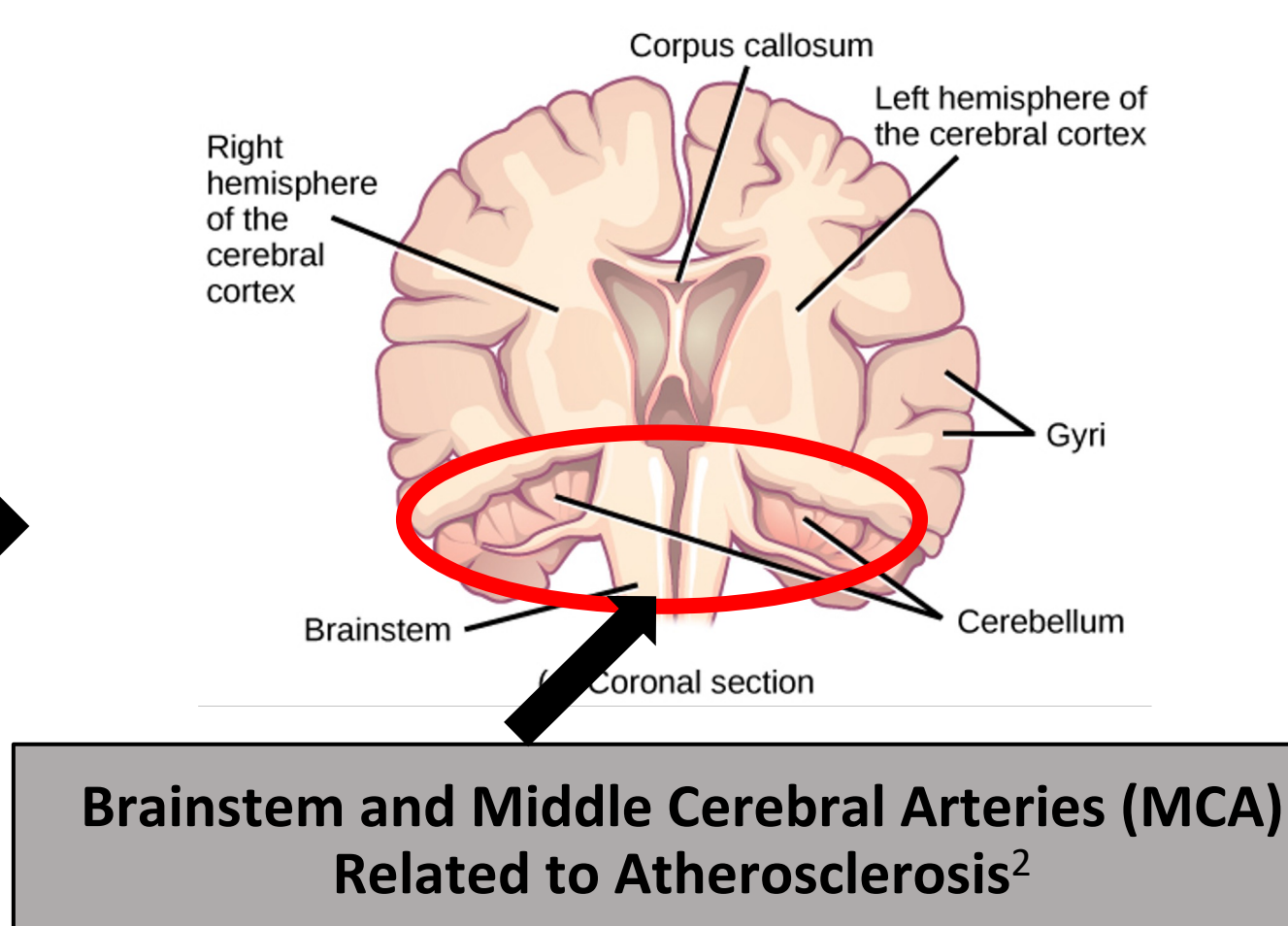
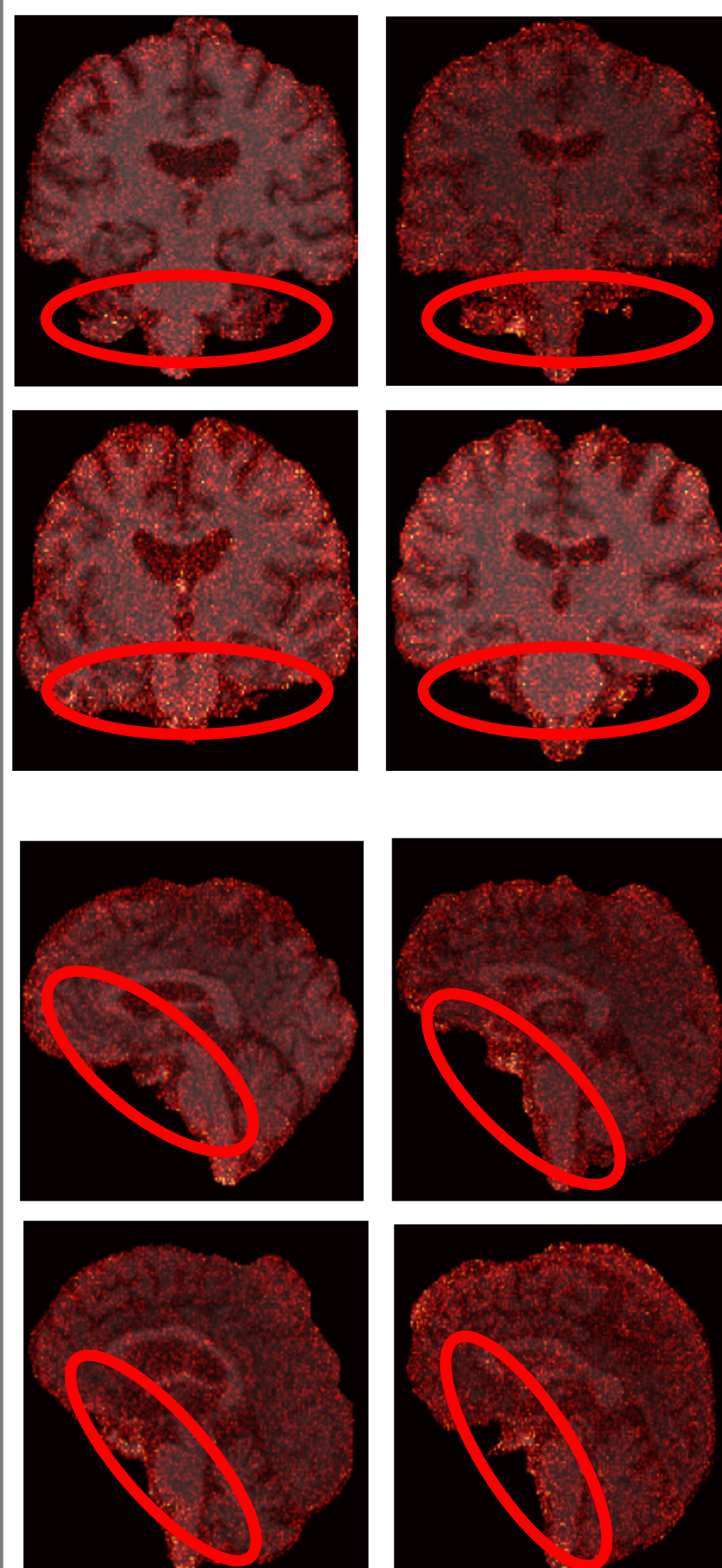
Result

Model Performance

Final Model with Matched Dataset (after optimization)
Metrics calculated with testing dataset
sensitivity:0.8
specificity:0.45



Heat Map Analysis



Result

- 95% Confidence Interval for AUROC is **[0.4176, 0.7917]**, which means AUROC could still be lower than 0.5
- At the optimal threshold of **0.4951**, the accuracy is **0.7**
- The highlighted areas are distributed throughout the brain, yet the **brainstem, MCA, and ACA** are especially noticeable

Conclusions

- This study demonstrates the **early feasibility** of classifying more diffuse pathologies of CVD using structural MRI scans
- Suitable pipeline for **biomarker detection** in neuroimaging, but not ready for clinical use
- Ongoing work includes constructing similar CNNs with MRI scans for the detection of other diseases, like **hypertension** and **atherosclerosis**

Limitations

- **Very limited dataset** (only **237** positive cases in the matched dataset)
- Restricted ability to detect **more specific CVDs**
- Some unknown **confounding variables** may influence the detection, like the comorbidity between CVD and other diseases
- MRI scans are not necessarily proximal to diagnosis (**prognosis or diagnosis?**)

Future Directions

- Transfer learning may be helpful
- Combine our dataset with other vascular outcomes to increase the dataset size
- Introduce nested cross-validation for improved generalizability
- Data augmentation

References and Acknowledgements

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